📄 DEVOPS SOP – Fusionpact Gauntlet Challenge

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🔗 GitHub Repository:

https://github.com/Vivek8076/fusionpact-devops-challenge

📌 Project Title:

Dockerized Full-Stack Application with CI/CD & Monitoring (Prometheus + Grafana)

1️⃣ Project Overview

This project is a complete **DevOps implementation** of a full-stack application consisting of a frontend service, backend API, and a monitoring stack.

The goal was to containerize services, automate deployment, and implement observability using industry-standard tools.

2️⃣ Architecture Design

Frontend (Nginx)

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Backend (FastAPI - Python)

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/metrics endpoint (Prometheus client)

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Prometheus (Data scraping)

↓

Grafana (Dashboard visualization)

3️⃣ Docker Setup

All services are containerized using Docker to ensure consistency across environments.

Backend Dockerfile

```
FROM python:3.10

WORKDIR /app

COPY requirements.txt .
RUN pip install -r requirements.txt

COPY . .

EXPOSE 8000

CMD ["uvicorn", "app.main:app", "--host", "0.0.0.0", "--port", "8000"]
```

Frontend (Nginx container)

- Static frontend served using Nginx
- Port mapping: 3000 → 80

4️⃣ Docker Compose Orchestration

```
version: "3.8"

services:

backend:
build: ./backend
container_name: backend_app
ports:
- "8000:8000"
restart: always

frontend:
build: ./frontend
container_name: frontend_app
ports:
- "3000:80"
depends_on:
- backend
restart: always

prometheus:
image: prom/prometheus
container_name: prometheus
volumes:
- ./prometheus.yml:/etc/prometheus/prometheus.yml
ports:
- "9090:9090"
restart: always

grafana:
```

```
image: grafana/grafana
container_name: grafana
ports:
- "3001:3000"
restart: always
```

5️⃣ Monitoring Setup (Prometheus + Grafana)

Backend Metrics Implementation

A custom Prometheus metric was added to track request counts.

```
from fastapi import FastAPI, Response
from prometheus_client import Counter, generate_latest

app = FastAPI()

REQUEST_COUNT = Counter("app_requests_total", "Total API Requests")

@app.get("/")
def home():
REQUEST_COUNT.inc()
return {"message": "API is running successfully"}

@app.get("/metrics")
def metrics():
return Response(generate_latest(), media_type="text/plain")
```

Prometheus Configuration

```
global:
scrape_interval: 5s

scrape_configs:
- job_name: "backend_service"
static_configs:
- targets: ["backend:8000"]
```

6️⃣ CI/CD Pipeline (GitHub Actions)

A simple CI pipeline was configured for automated build testing.

```
name: CI Pipeline

on:
push:
branches: [ main ]

jobs:
build:
runs-on: ubuntu-latest

steps:
- name: Checkout Repository
uses: actions/checkout@v3

- name: Build Docker Images
run: docker-compose build
```

```
- name: Start Containers
run: docker-compose up -d
```

7️⃣ Data Persistence Strategy

Although the application is containerized and stateless, persistence is ensured through:

- Docker volume isolation for service independence
- Prometheus local storage for metrics retention
- Grafana dashboard configuration stored in container volumes
- Backend designed to be stateless for scalability

8️⃣ Execution Commands

```
docker-compose up --build
docker ps
docker logs backend_app
git add .
git commit -m "Added full DevOps stack with monitoring and CI/CD"
git push origin main
```

9️⃣ Verification Steps

Backend API

http://localhost:8000

Metrics Endpoint

http://localhost:8000/metrics

Prometheus Dashboard

http://localhost:9090

Grafana Dashboard

http://localhost:3001

Login:

admin / admin

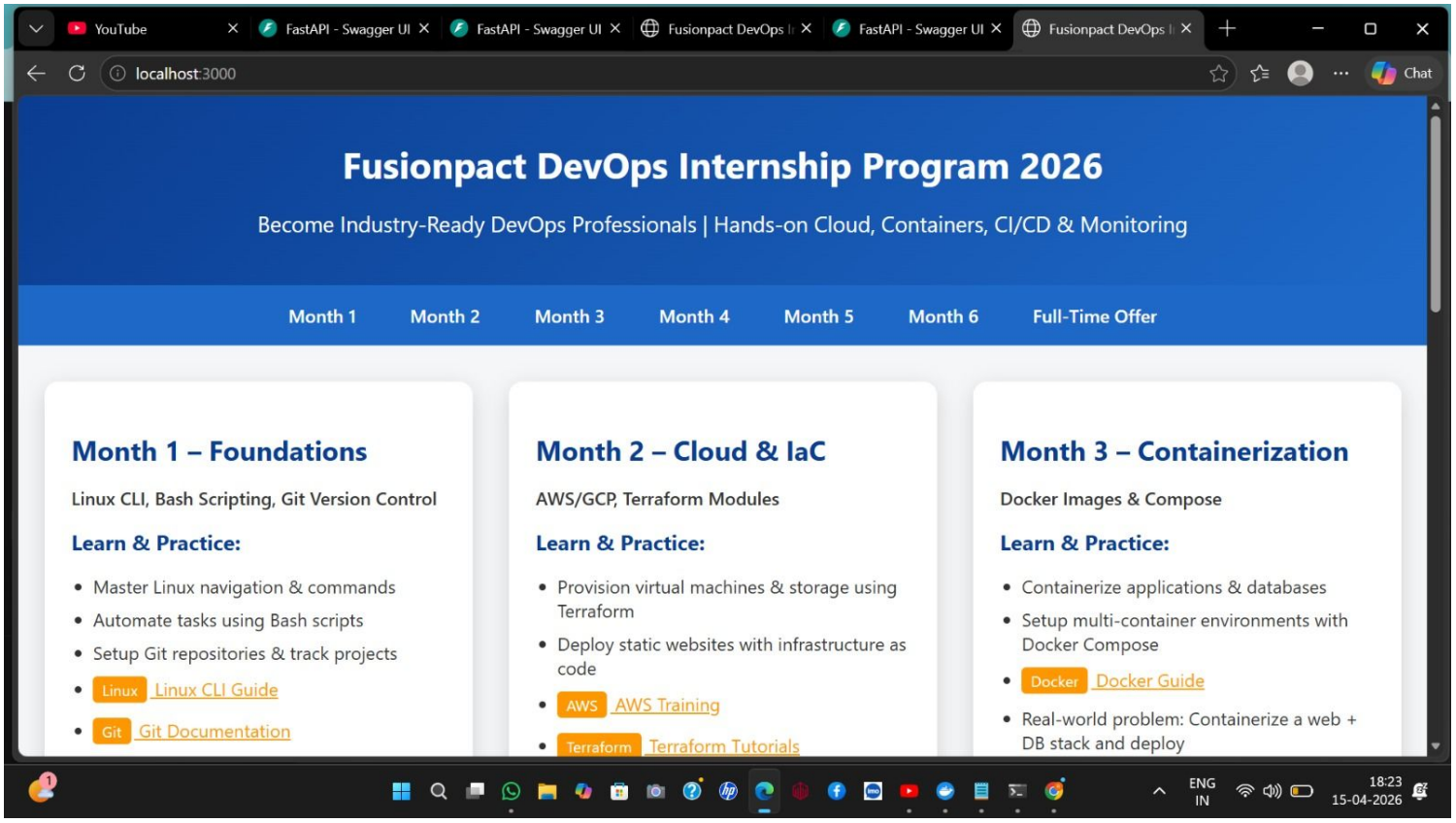
🔟 Key Learnings

- Containerization using Docker
- Multi-service orchestration using Docker Compose
- Observability using Prometheus + Grafana
- CI/CD automation using GitHub Actions
- Real-world DevOps workflow understanding

🏁 Final Summary

This project demonstrates a production-style DevOps pipeline with full-stack integration, monitoring, and automation.

The system is scalable, containerized, and ready for deployment in a real-world environment.


YouTube
FastAPI - Swagger UI
FastAPI - Swagger UI
Fusionpact DevOps I
FastAPI - Swagger UI
Fusionpact DevOps I
localhost:3000
Chat
Fusionpact DevOps Internship Program 2026
Become Industry-Ready DevOps Professionals | Hands-on Cloud, Containers, CI/CD & Monitoring
Month 1
Month 2
Month 3
Month 4
Month 5
Month 6
Full-Time Offer
Month 1 – Foundations
Linux CLI, Bash Scripting, Git Version Control
Learn & Practice:
Master Linux navigation & commands
Automate tasks using Bash scripts
Setup Git repositories & track projects
Linux Linux CLI Guide
Git Git Documentation
Month 2 – Cloud & IaC
AWS/GCP, Terraform Modules
Learn & Practice:
Provision virtual machines & storage using Terraform
Deploy static websites with infrastructure as code
AWS AWS Training
Terraform Terraform Tutorials
Month 3 – Containerization
Docker Images & Compose
Learn & Practice:
Containerize applications & databases
Setup multi-container environments with Docker Compose
Docker Docker Guide
Real-world problem: Containerize a web + DB stack and deploy
ENG
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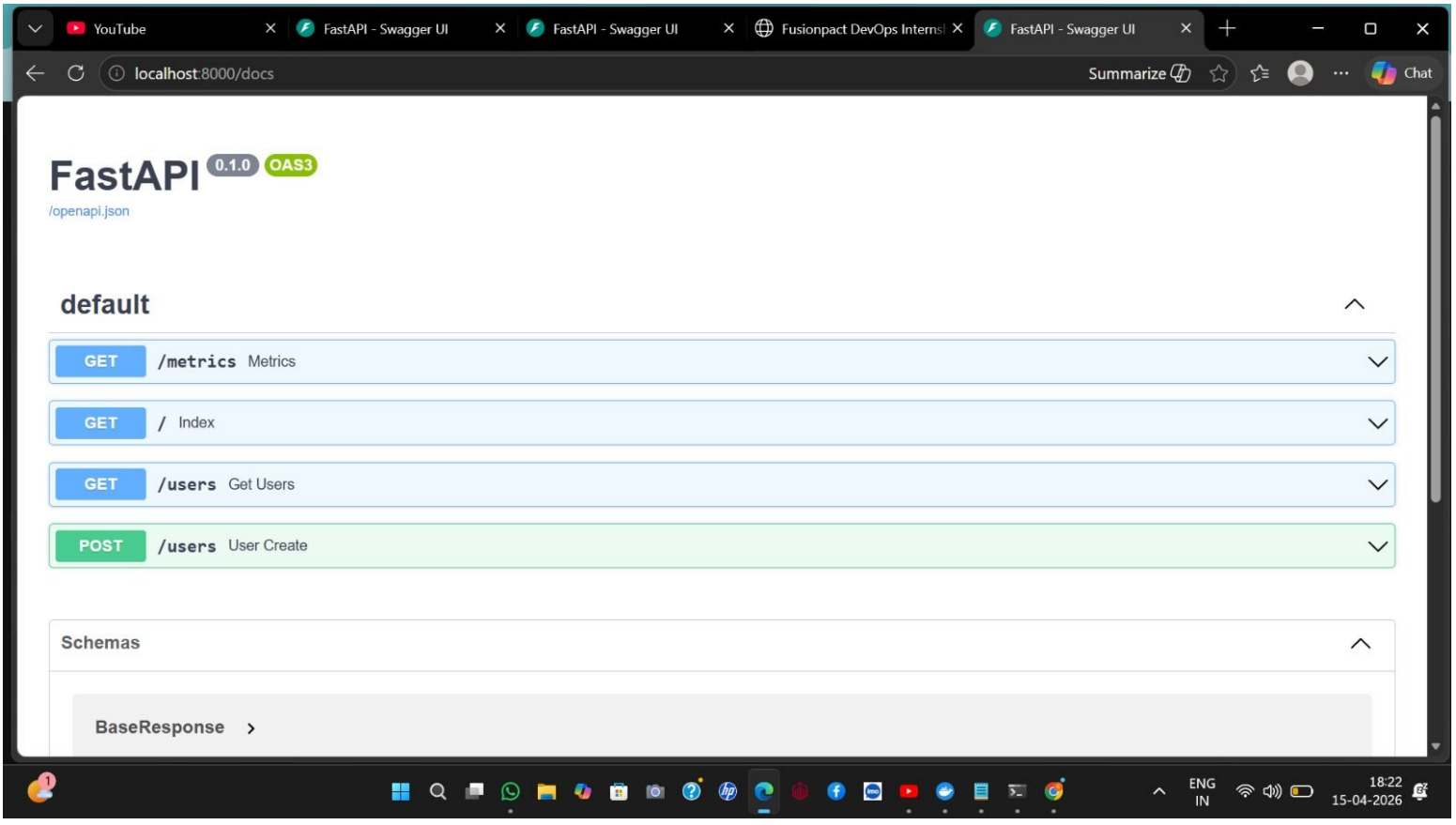
18:23
15-04-2026


YouTube
FastAPI - Swagger UI
FastAPI - Swagger UI
Fusionpact DevOps Intern
FastAPI - Swagger UI
localhost:8000/docs
Summarize
Chat
FastAPI
0.1.0
OAS3
/openapi.json
default
GET /metrics Metrics
GET / Index
GET /users Get Users
POST /users User Create
Schemas
BaseResponse
ENG
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Docker Running Containers

```
C:\Users\HP\fusionpact-devops-challenge\backend>docker ps
CONTAINER ID   IMAGE                                   COMMAND                  CREATED          STATUS              PORTS
                                   NAMES
8fe074046959   fusionpact-devops-challenge-frontend   "/docker-entrypoint.…"   20 minutes ago   Up About a minute   0.0.0.0:3000->80/
tcp, [::]:3000->80/tcp       frontend_app
7095caba2f0c   prom/prometheus                         "/bin/prometheus --c…"   20 minutes ago   Up About a minute   0.0.0.0:9090->909
0/tcp, [::]:9090->9090/tcp   prometheus
46847eed044c   grafana/grafana                         "/run.sh"                20 minutes ago   Up About a minute   0.0.0.0:3001->300
0/tcp, [::]:3001->3000/tcp   grafana
376c11064585   fusionpact-devops-challenge-backend    "uvicorn app.main:ap…"   20 minutes ago   Up About a minute   0.0.0.0:8000->800
0/tcp, [::]:8000->8000/tcp   backend_app

C:\Users\HP\fusionpact-devops-challenge\backend>
```

```
C:\Users\HP\fusionpact-devops-challenge\frontend>docker ps
CONTAINER ID   IMAGE                                   COMMAND                  CREATED          STATUS          PORTS
                                   NAMES
8fe074046959   fusionpact-devops-challenge-frontend   "/docker-entrypoint.…"   22 minutes ago   Up 3 minutes    0.0.0.0:3000->80/tcp,
[::]:3000->80/tcp       frontend_app
7095caba2f0c   prom/prometheus                         "/bin/prometheus --c…"   22 minutes ago   Up 3 minutes    0.0.0.0:9090->9090/tcp
, [::]:9090->9090/tcp   prometheus
46847eed044c   grafana/grafana                         "/run.sh"                22 minutes ago   Up 3 minutes    0.0.0.0:3001->3000/tcp
, [::]:3001->3000/tcp   grafana
376c11064585   fusionpact-devops-challenge-backend    "uvicorn app.main:ap…"   22 minutes ago   Up 3 minutes    0.0.0.0:8000->8000/tcp
, [::]:8000->8000/tcp   backend_app

C:\Users\HP\fusionpact-devops-challenge\frontend>
```

Command Execution

```
Command Prompt

C:\Users\HP\fusionpact-devops-challenge>dir
 Volume in drive C is Windows
 Volume Serial Number is 32A5-368C

 Directory of C:\Users\HP\fusionpact-devops-challenge

15-04-2026  22:30    <DIR>          .
15-04-2026  17:49    <DIR>          ..
15-04-2026  21:25    <DIR>          .github
15-04-2026  17:49                45 .gitignore
15-04-2026  18:06    <DIR>          backend
15-04-2026  22:12               551 docker-compose.yml
15-04-2026  18:14    <DIR>          frontend
15-04-2026  22:30               130 prometheus.yml
15-04-2026  17:49             4,924 README.md
15-04-2026  17:49           754,629 SOP CREATE HOME WEBPAGE USING NGINX SERVER.pdf
               5 File(s)        760,279 bytes
               5 Dir(s)   3,732,557,824 bytes free

C:\Users\HP\fusionpact-devops-challenge>
```

```
Command Prompt

15-04-2026  17:49    <DIR>          ..
15-04-2026  21:25    <DIR>          .github
15-04-2026  17:49                45 .gitignore
15-04-2026  18:06    <DIR>          backend
15-04-2026  21:53                 0 docker-compose.yml
15-04-2026  18:14    <DIR>          frontend
15-04-2026  17:49             4,924 README.md
15-04-2026  17:49           754,629 SOP CREATE HOME WEBPAGE USING NGINX SERVER.pdf
               4 File(s)        759,598 bytes
               5 Dir(s)   4,914,634,752 bytes free

C:\Users\HP\fusionpact-devops-challenge>notepad docker-compose.yml

C:\Users\HP\fusionpact-devops-challenge>dir
 Volume in drive C is Windows
 Volume Serial Number is 32A5-368C

 Directory of C:\Users\HP\fusionpact-devops-challenge

15-04-2026  21:53    <DIR>          .
15-04-2026  17:49    <DIR>          ..
15-04-2026  21:25    <DIR>          .github
15-04-2026  17:49                45 .gitignore
15-04-2026  18:06    <DIR>          backend
15-04-2026  21:56               268 docker-compose.yml
15-04-2026  18:14    <DIR>          frontend
15-04-2026  17:49             4,924 README.md
15-04-2026  17:49           754,629 SOP CREATE HOME WEBPAGE USING NGINX SERVER.pdf
               4 File(s)        759,866 bytes
               5 Dir(s)   4,913,614,848 bytes free

C:\Users\HP\fusionpact-devops-challenge>
```

Command Execution

Command Prompt - docker-c

```
frontend_app  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
frontend_app  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
frontend_app  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
frontend_app  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
frontend_app  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
frontend_app  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
frontend_app  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
frontend_app  | /docker-entrypoint.sh: Configuration complete; ready for start up
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: using the "epoll" event method
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: nginx/1.29.8
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: built by gcc 15.2.0 (Alpine 15.2.0)
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker processes
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 30
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 31
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 32
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 33
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 34
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 35
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 36
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 37
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 38
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 39
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 40
frontend_app  | 2026/04/15 16:27:55 [notice] 1#1: start worker process 41
backend_app   | INFO:     Started server process [1]
backend_app   | INFO:     Waiting for application startup.
backend_app   | INFO:     Application startup complete.
backend_app   | INFO:     Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)

v View in Docker Desktop   o View Config   w Enable Watch   d Detach
```

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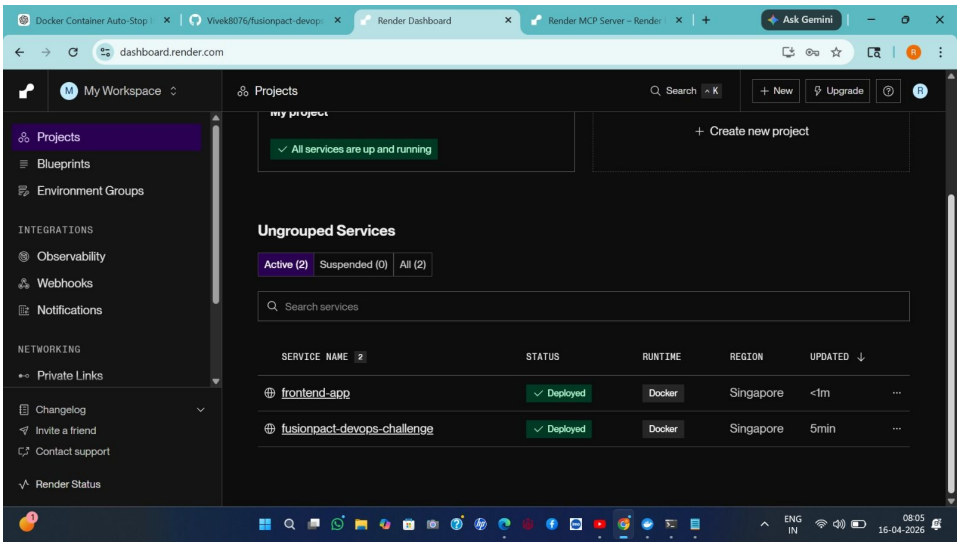
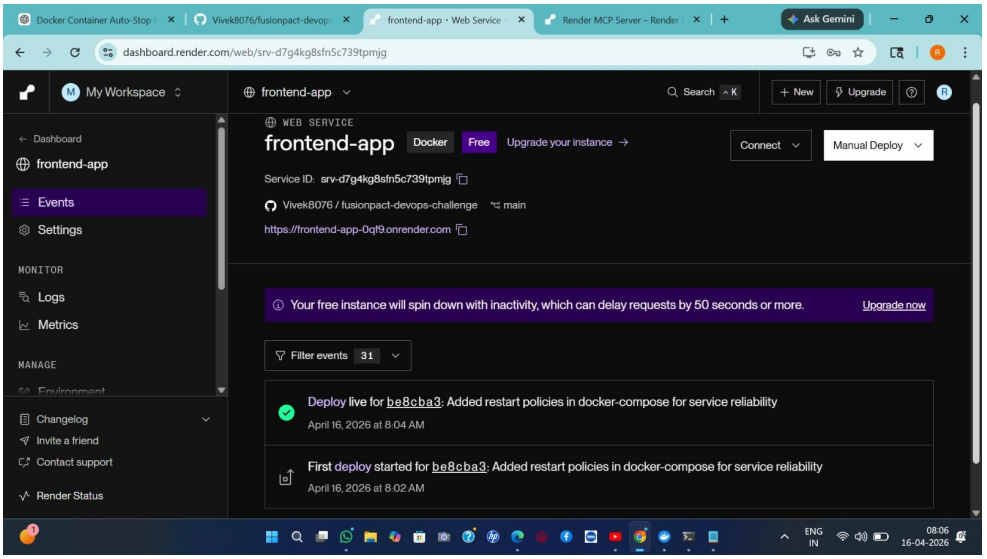
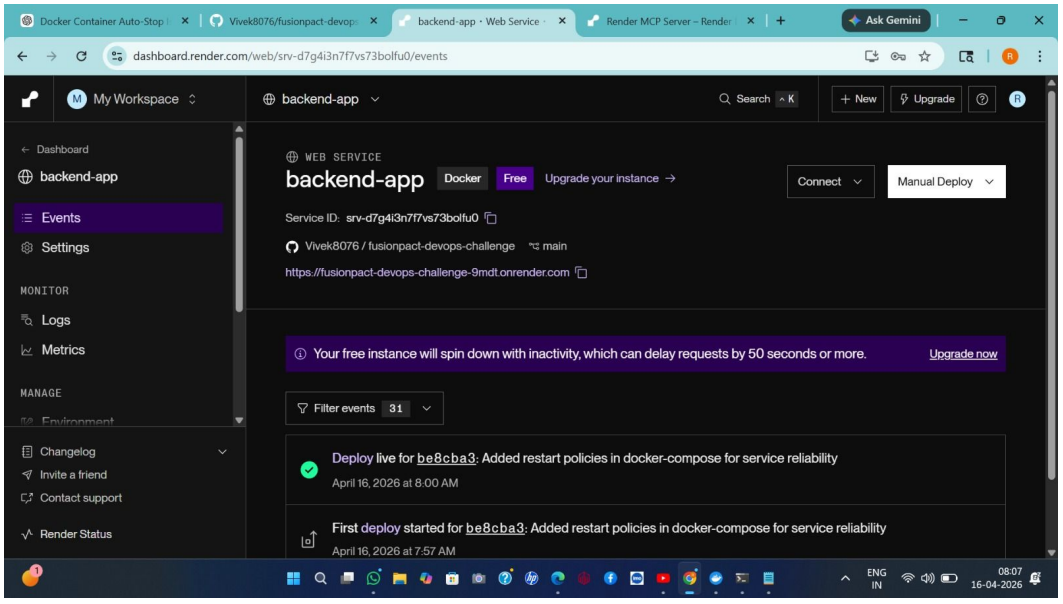
Screenshots

The following screenshots demonstrate the successful local execution and cloud deployment of the application

Render Dashboard



Frontend (Live Application)



⚙ Backend (API Service)